

Vector Databases and Embeddings

What are Embeddings?

An embedding is a numerical representation of information such as text. Text with similar meaning tends to have vectors that are close together in an embedding space. Embeddings make semantic search possible because systems can compare numerical representations rather than relying only on exact keyword matches.

Vector Databases

A vector database stores embeddings and provides efficient similarity search. Examples include Qdrant, Pinecone, Weaviate, and Chroma. In a RAG application, document chunks and their embeddings are stored in a vector database so that relevant information can be retrieved when a user submits a query.

Similarity Search

A query can also be converted into an embedding. The vector database compares the query vector with stored document vectors and returns the most similar results. Common similarity measures include cosine similarity, dot product, and Euclidean distance.

Metadata

Vector records can contain metadata such as the original filename, page number, document category, author, or creation date. Metadata can be used to filter search results and to identify the source of retrieved information.